

Rohith Pudari

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EDUCATION

University of Toronto Fall 2022 – Nov 2026 (Expected)

Ph.D., Dept of Electrical & Computer Engineering, Advisor: Dr. Shurui Zhou

University of Victoria Sep 2020 – Aug 2022

M.Sc., Computer Science, Advisor: Dr. Neil Ernst

Thesis: *AI Supported Software Development: Moving beyond Code Completion*

Jawaharlal Nehru Technological University Jul 2015 – Aug 2019

B.Tech., Information Technology

INDUSTRY EXPERIENCE

Data Scientist May 2019 – Aug 2020

Deloitte TS, India

- Developed a custom AI model for a Fortune 500 pharma client to predict product demand and risk factors, improving forecast accuracy by 17%.
- Implemented ensemble modeling and real-time monitoring pipelines with visualization dashboards, enhancing decision-making speed and reliability.
- Created architecture for a distributed server cluster supporting global data access and analytics.

Information Technology Intern Jan 2019 – May 2019

Deloitte TS, India

- Implemented an application where risks associated with a product are calculated using custom questionnaires thereby providing risk mitigation plans.
- Created tests and documentation for an internal tool to detect code smells and test coverage reports.

Software Engineering Intern Jul. 2018 – Sep 2018

Google KA, India

- Created the visual flow builder for the DialogueFlow web interface.
- Worked on creating Dialogue-flow intent generation process.
- Created a CI/CD pipeline for scaling the payment gateway for google pay application.
- Gave multiple talks on the usages and ways to integrate DialogueFlow into existing softwares at conferences and meetups.

SELECTED PUBLICATIONS

- **R. Pudari**, A. Awon Musa, N. Ernst, S. Zhou, "*When Tools Overlook Domain Knowledge: An Empirical Study of Refactoring in Scientific Software*," ACM Transactions on Software Engineering and Methodology (TOSEM) 2026,
- **R. Pudari**, S. Zhou, I. Ahmed, Z. Dai and S. Zhou, "*Aligning Documentation and Q&A Forum through Constrained Decoding with Weak Supervision*," 2023 IEEE International Conference on Software Maintenance and Evolution (ICSME), Bogotá, Colombia, 2023, pp. 346-351, doi: 10.1109/ICSME58846.2023.00043. [LINK].
- **R. Pudari**, N. Ernst, "*From Copilot to Pilot: Toward AI Supported Software Engineering*". arXiv preprint arXiv:2303.04142 (2023) [LINK].

TEACHING EXPERIENCE

University of Toronto:

- Head Teaching Assistant - ECE1785/CSC2130: Empirical Software Engineering (Spring 2024, 2025)
- Head Teaching Assistant - ECE444: Software Engineering (Fall 2022, 2023, 2024, 2025)
- Teaching Assistant - ECE324: Machine Intelligence, Software and Neural Networks (Spring 2026)
- Teaching Assistant - ECE344: Operating Systems (Fall 2025)
- Teaching Assistant - CSC209: Software Tools and Systems Programming (Spring 2023)

University of Victoria:

- Teaching Assistant - SENG 321: Requirements Engineering (Spring 2021, 2022)
- Teaching Assistant - SENG 275 - Software Testing (Summer 2021)
- Teaching Assistant - SENG 350 - Software Architecture and Design (Fall 2021)
- Teaching Assistant - SENG 330 - Object-Oriented Software Development (Fall 2020)

NOTABLE OPEN SOURCE CONTRIBUTIONS

- **Keras:** Popular repository sponsored by google with a multi-backend approach to build and train DL models with JAX, TensorFlow, and PyTorch seamlessly.
 - PR#21566 - Added daily Python CPU-only tests to nightly workflow, improving the CI/CD pipeline to automate testing on multiple python versions.
 - PR#20447 - Added a new function (IFFT2) that computes the 2-dimensional inverse discrete Fourier Transform.
- **PySyft:** Package to perform data science on data that remains in someone else's server in a privacy preserving way.
 - PR#5649 - Added scripts to check install and autoupdate a dependency to the latest supported version.

INVITED TALKS

- Pluriza Webinar, "Reasons why ChatBots need Natural language processing," December 2022.
- BV Raju Institute of Technology CSE Departmental Seminar, "Introduction to Deep Learning with Pytorch and use cases", March 2021.
- Startup Masterclass at IIT Kanpur, "NLP Techniques to use documentations in their knowledge base without fine-tuning", June 2020.

TECHNICAL SKILLS

- *Computing Languages:* Python, C, SQL, Java, Javascript, C++.
- *Tools:* TensorFlow, Keras, Pytorch, Scikit-learn, NumPy, SciPy, NLTK, SpaCy, Gensim, AllenNLP, XGBoost, LibSVM, SVMlight.